Dataset: Sales_dataset

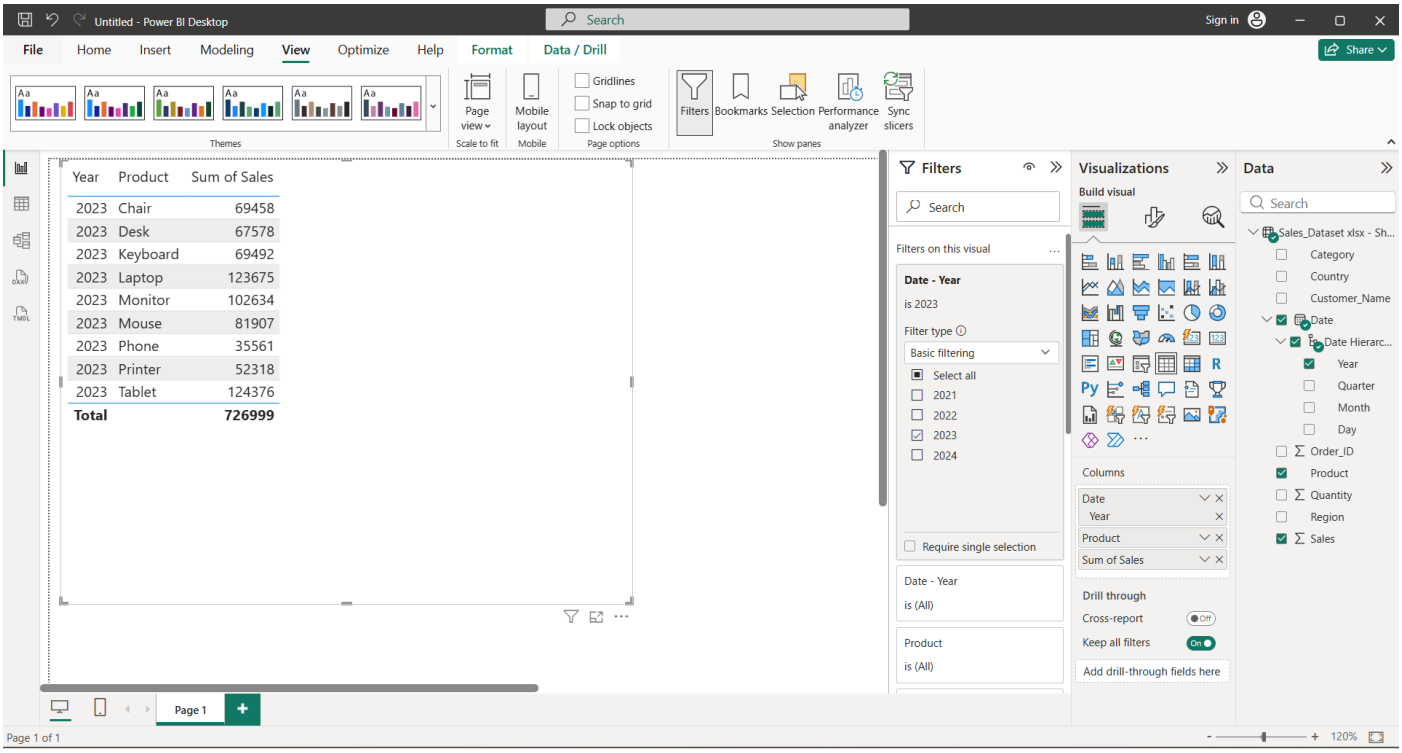
1. **What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI ?**

Ans:-

Feature	Basic Filte	Advanced Filter	Slicer
Definition	Simple filter to select values from a list	Filter using conditions (rules)	Visual filter placed on report
Type	Value-based	Condition-based	Interactive UI element
Location	Filter Pane	Filter Pane	Report Canvas
How it works	Select/deselect checkboxes	Apply logic (AND / OR)	Click buttons, dropdown, or list
User Interaction	Limited	Limited	High (interactive)
Filtering Logic	Exact values only	Custom conditions (>, <, contains, etc.)	Same as basic but visual
Example	Select Region = East, West	Sales > 20000 AND < 50000	Click “Electronics” category
Best Use Case	Quick filtering	Complex filtering conditions	Dashboard user interaction

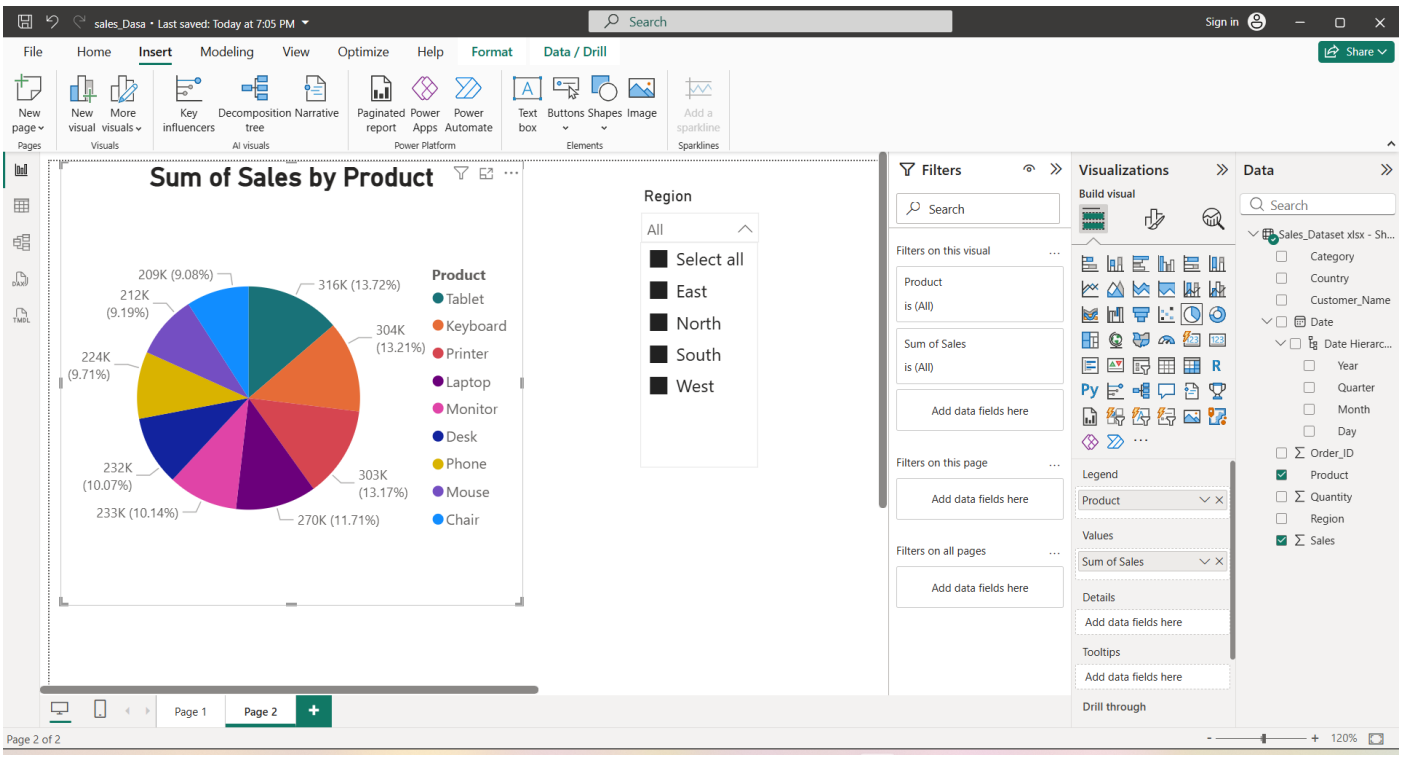
2. **Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.**

Ans:-



3. **Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region.**

Ans:-



4. **What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?**

Ans:-

Visual-Level Filter:-

A Visual-Level Filter is a filter that is applied to only one specific visual (chart, table, graph).

It affects only that selected visual, not the rest of the report.

Example:
If you apply a filter Year = 2023 on a pie chart, only that pie chart will show 2023 data, while other visuals can still show all years.

Difference Between Filter Types

1. Visual-Level Filter

- Applies to: **Single visual only**
- Location: *Filters on this visual*
- Effect: Does NOT impact other visuals

2. Page-Level Filter

- Applies to: All visuals on one page

- Location: *Filters on this page*
- Effect: Every chart/table on that page is filtered

Example:
Filter Region = East → all visuals on the page show only East data

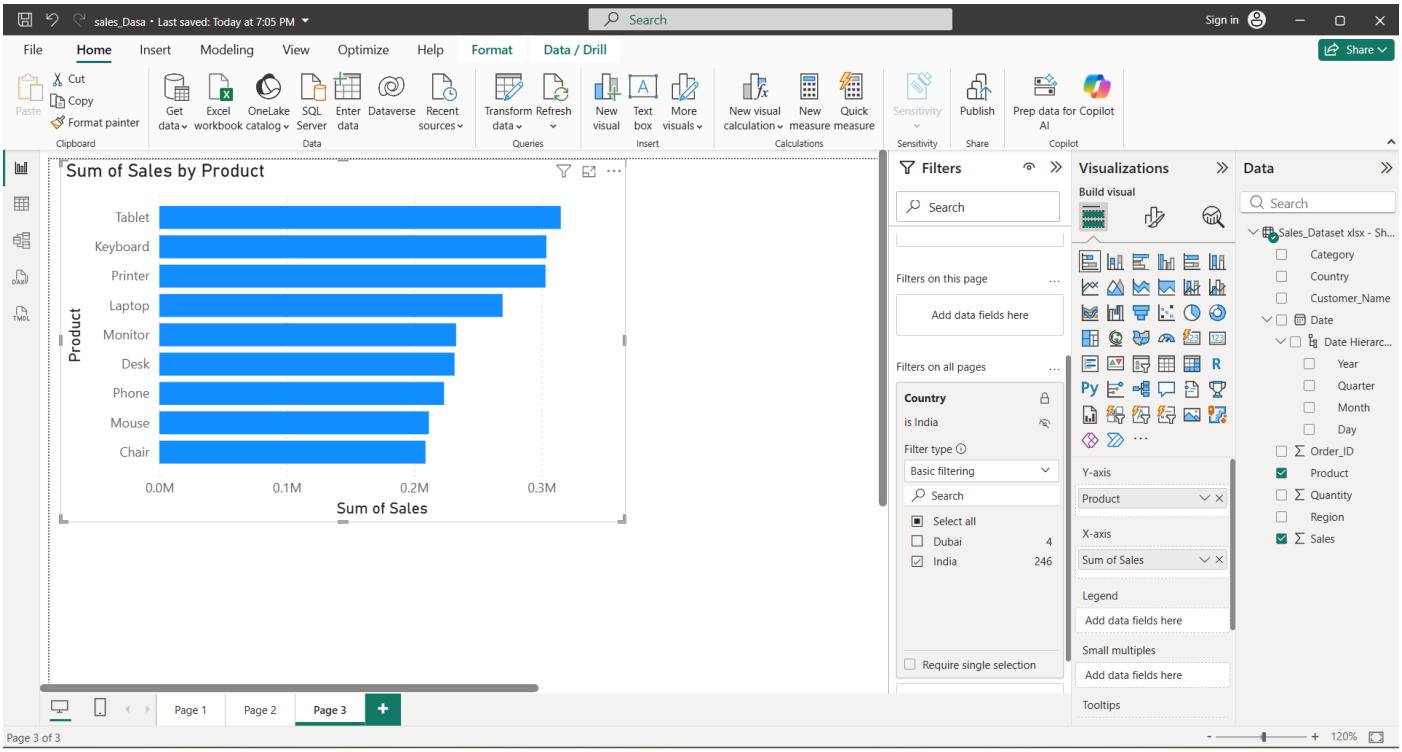
3. Report-Level Filter

- Applies to: Entire report (all pages)
- Location: *Filters on all pages*
- Effect: Affects every visual across all pages

Example:
Filter Year = 2023 → entire report shows only 2023 data

5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.

Ans:-



6. **What is the role of Drill-Down in Power BI? How does it improve data exploration?**

Ans:-

Role of Drill-Down:-

Drill-Down is a feature that allows users to navigate from higher-level data to more detailed levels within a visual.

It works using a hierarchy (e.g., Year → Quarter → Month → Day).

Example

If you have a chart showing:

- Year-wise sales

Using Drill-Down, you can click and go deeper into:

- Year → Quarter → Month → Day

Role of Drill-Down

- Helps users analyze data in detail
- Breaks down summarized data into granular levels
- Makes reports interactive and dynamic

How It Improves Data Exploration

1. Deeper Insights

- Users can explore hidden details behind summary data

2. Interactive Analysis

- Instead of static reports, users can click and explore

3. Better Decision Making

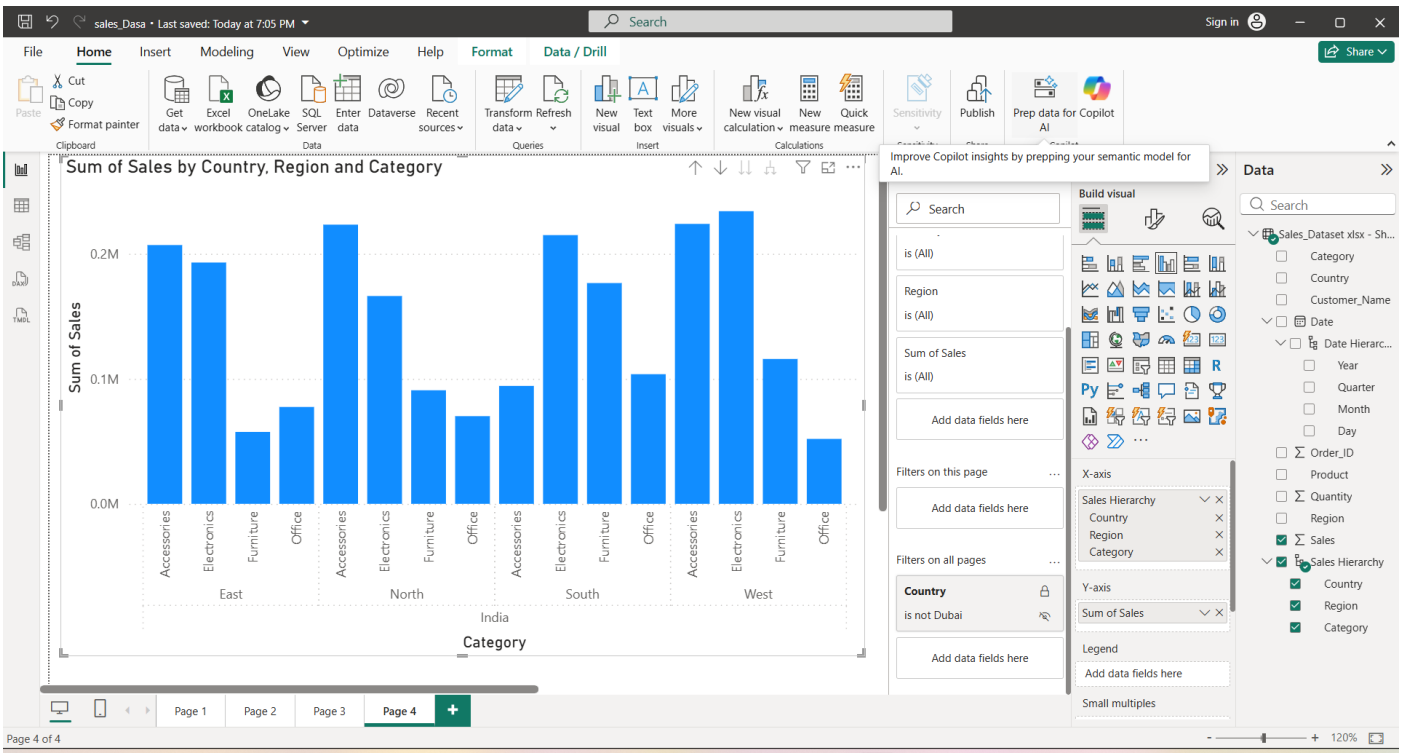
- Detailed data helps in making accurate decisions

4. Easy Navigation

- Move between levels without creating multiple visuals

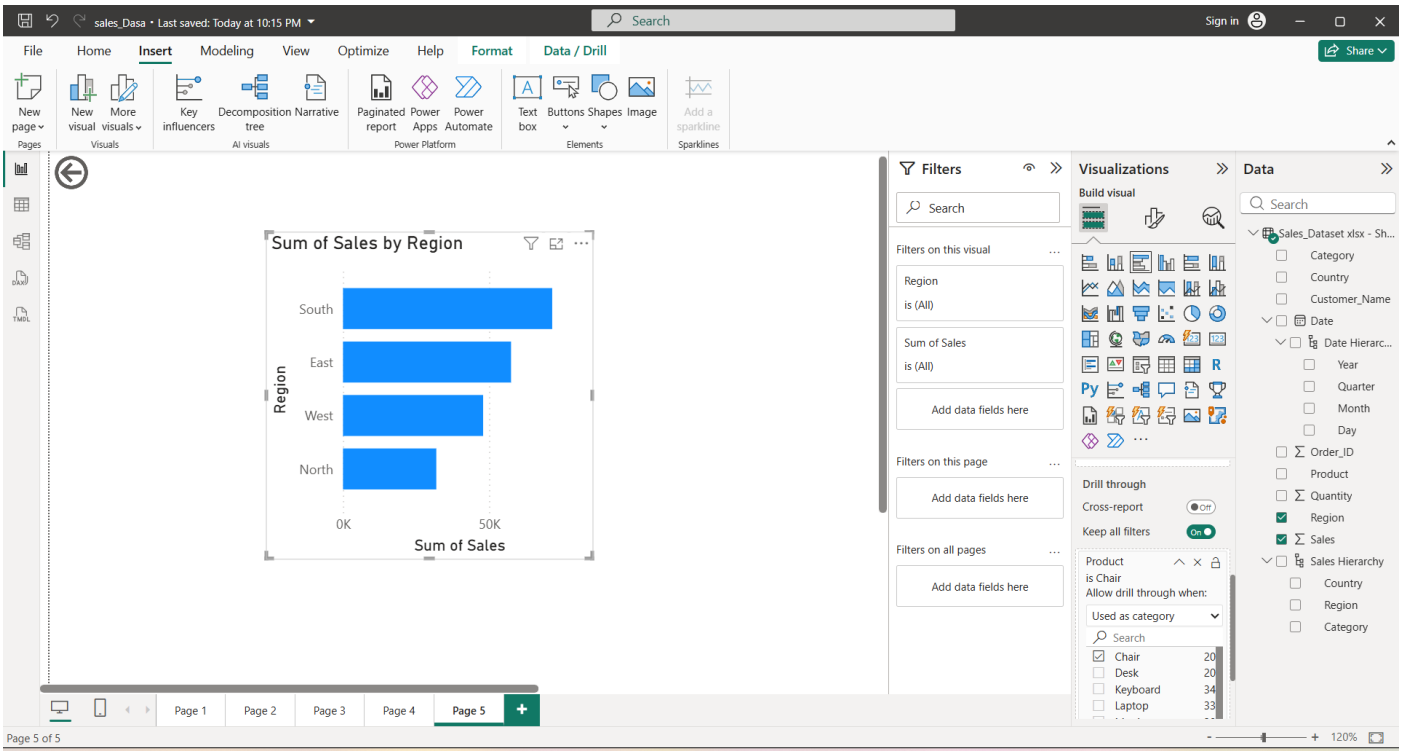
7. **Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.**

Ans:-



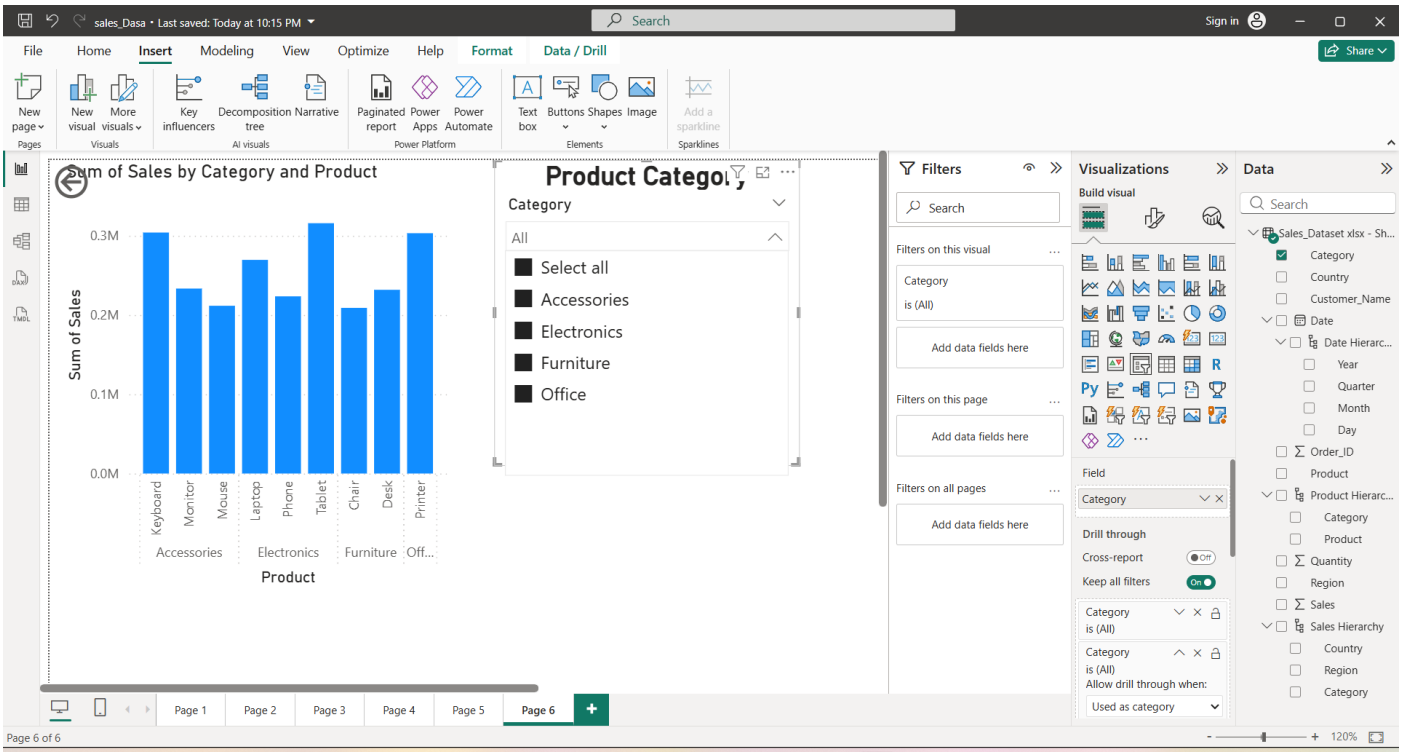
8. **Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region-wise sales.**

Ans:-



9. **Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy**

Ans:-



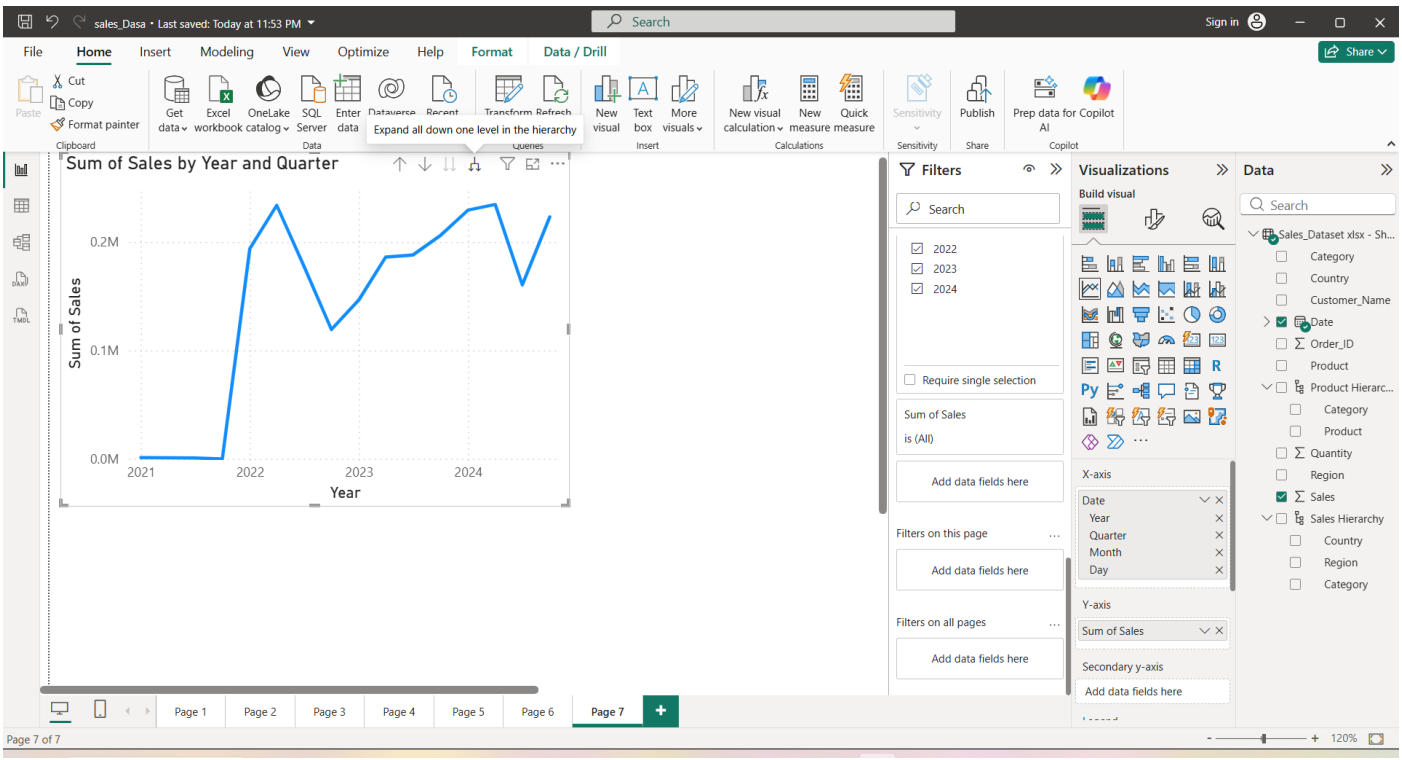
10. **Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.**

Ans:-

Feature	Drill-Down	Drill-Through
Definition	Allows users to explore data within the same visual by going to a lower level in a hierarchy	Allows users to navigate to a different report page with more detailed data
Navigation Type	Stays on the same chart/page	Moves to a new page
Data Structure	Requires a hierarchy (e.g., Category → Product)	Works using filters passed to another page
User Action	Click on bars + use drill icons (⬇)	Right-click on a data point → select drill-through
Level of Detail	Goes deeper step-by-step in levels	Shows full detailed report for selected item
Setup Complexity	Simple (just create hierarchy)	Requires creating a separate drill-through page
Use Case	Quick data exploration	Detailed analysis/reporting
Example	You click on Electronics → see Products (Laptop, Phone)	You right-click Laptop → go to detailed page
Purpose	Explore hierarchy quickly	Analyze one item in depth

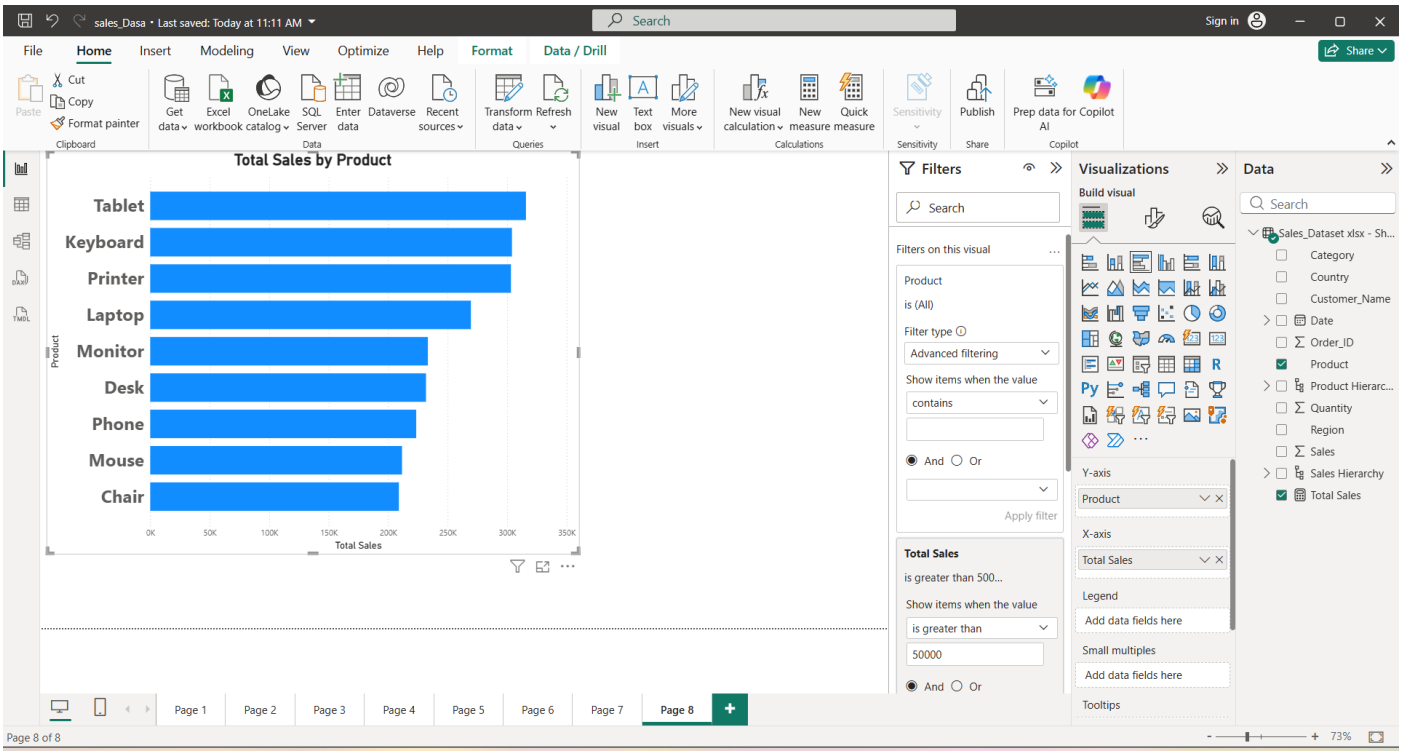
11.Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.

Ans:-



12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000.Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.

Ans:-



13. What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction.

Ans:-

Keep All Filters in Drill-Through (Power BI)

Keep All Filters:-

“Keep All Filters” is an option in Drill-Through that ensures all filters applied on the source page (such as slicers, page filters, and visual filters) are carried over to the target drill-through page.

Options in Drill-Through (Power BI)

In **Drill-Through**, there are a few important options :-

Drill-through Field:- The field used to pass context (e.g., Product, Category) to the target page.

Keep All Filters:- Keeps all filters (slicers, page filters, visual filters) when navigating to the drill-through page.

Cross-report:- Allows drill-through between different reports.

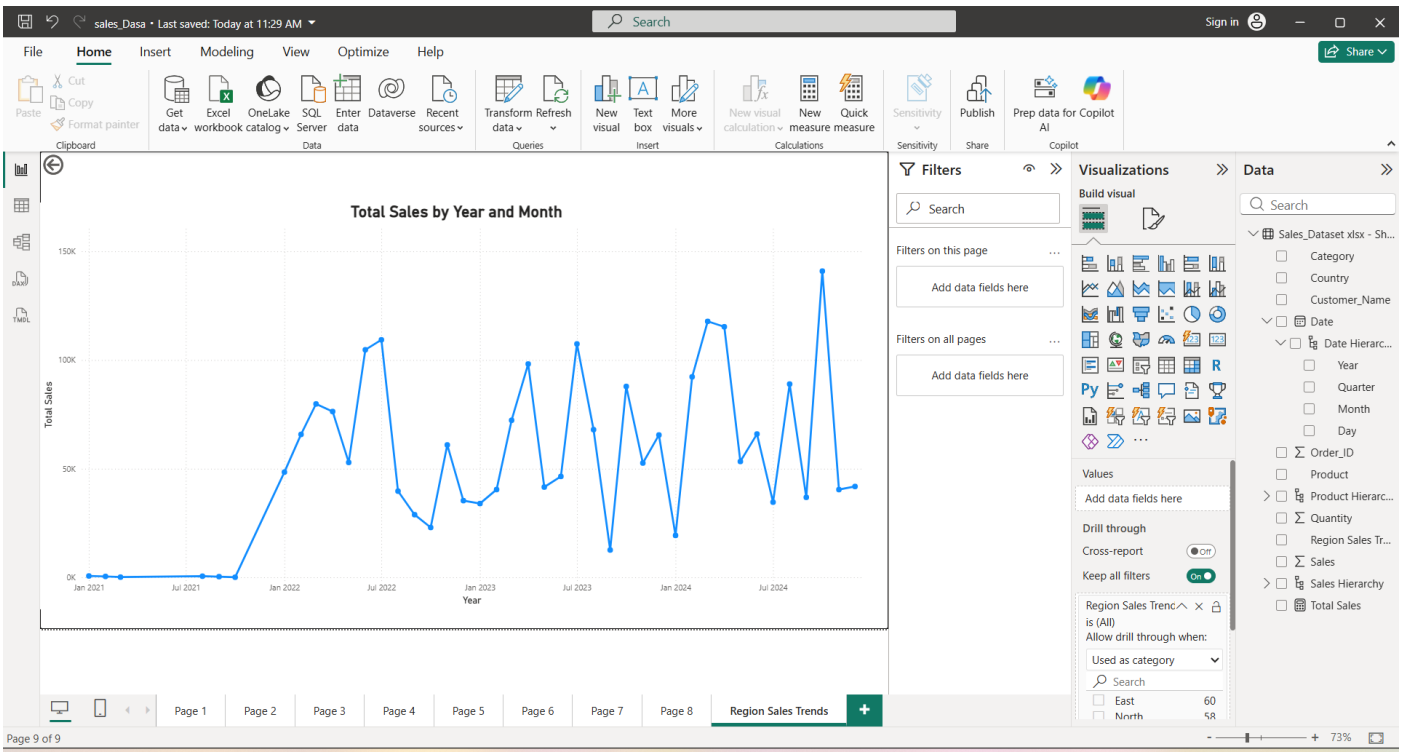
Keep all filters (Toggle):- Can be turned ON/OFF to control filter behavior.

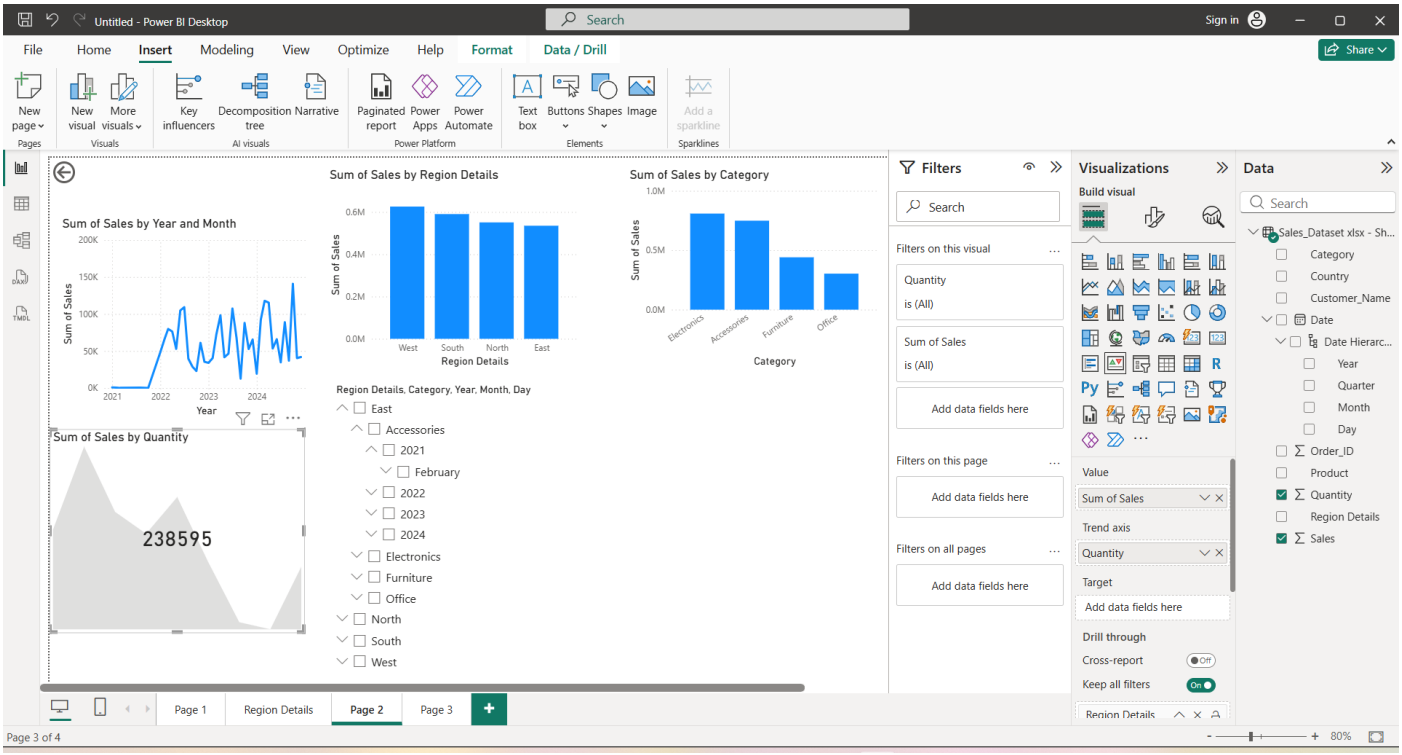
How It Affects User Interaction

Aspect	With “Keep All Filters” ON	With “Keep All Filters” OFF
Filter Transfer	All filters (slicer, page, visual) are passed	Only selected field (drill-through field) is passed
Data Context	Maintains full context of user selection	Limited context (only selected item)
User Experience	More detailed and relevant view	More general view
Analysis	Focused and consistent analysis	May require reapplying filters
Example	If user selects *Electronics* + *2023*, both apply on next page	Only *Electronics* passes, not 2023

14. Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.

Ans:-



15. Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data.

Ans:-